

Transforming Nonlinear Data with Logarithms

Certain relationships between two variables do not run in a straight line. Populations grow exponentially. Radioactive substances decay exponentially. When you try to fit a straight line to data like this, you may get a regression line, but the *line of best fit* will be wrong or non-existent. Logarithm models can straighten out curved relationships, allowing you to use linear regression techniques on data that is not linear in its raw form.

1.1 Exponential Models

A content creator posts a video and tracks how many views it gets each day:

Day (x)	0	1	2	3	4	5	6
Views, thousands (y)	4.2	8.3	17.4	34.1	77.8	161.8	337.5

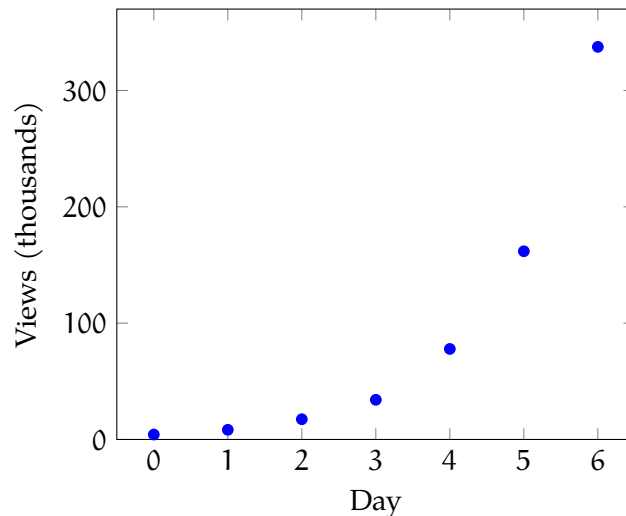


Figure 1.1: Daily views after posting. The curve bends upward instead of running straight.

A straight line fit to this data anyway would give r^2 around 0.90.

This would follow an exponential model, $y = a \cdot b^x$, where a is the starting count and b

is the daily growth factor. Therefore:

$$\log(y) = \log(a \cdot b^x) = \log(a) + x \log(b)$$

$\log(a)$ and $\log(b)$ are both constants, so the right side is linear in x , with slope $\log(b)$ and intercept $\log(a)$. Graph $\log(y)$ against x instead of y against x , and a genuine exponential relationship becomes a line. This is a *logarithmic transformation*: the data does not change, only the scale you graph it on.

Apply that here:

Day (x)	0	1	2	3	4	5	6
Views, thousands (y)	4.2	8.3	17.4	34.1	77.8	161.8	337.5
$\log_{10}(y)$	0.623	0.919	1.241	1.533	1.891	2.209	2.528

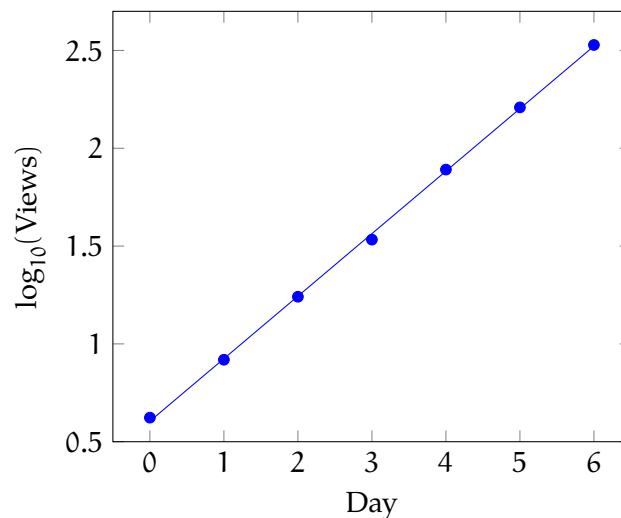


Figure 1.2: The same data after a logarithmic transformation. The points now fall along a line.

Computing in Python

Here is the video views example again, this time in Python:

```

1 import numpy as np
2 import statsmodels.api as sm
3
4 day = np.array([0, 1, 2, 3, 4, 5, 6])
5 views = np.array([4.2, 8.3, 17.4, 34.1, 77.8, 161.8, 337.5])
6
7 # Fit the raw model: views as a linear function of day

```

```

8 X = sm.add_constant(day)
9 raw_model = sm.OLS(views, X).fit()
10 print(raw_model.params)
11 print("R-squared:", raw_model.rsquared)
12 print(raw_model.resid)

```

This gives an intercept of -54.91 , a slope of 48.83 , and $R^2 \approx 0.750$.

Apply the log transform and refit:

```

1 # Apply a natural log transform to the response variable
2 log_views = np.log(views)
3
4 log_model = sm.OLS(log_views, X).fit()
5 print(log_model.params)
6 print("R-squared:", log_model.rsquared)
7 print(log_model.resid)

```

Now the intercept is near 1.393 , the slope near 0.7356 , and $R^2 \approx 0.9995$.

```

1 slope = log_model.params[1]
2 intercept = log_model.params[0]
3
4 growth_factor = np.exp(slope)
5 percent_growth = (growth_factor - 1) * 100
6 starting_views = np.exp(intercept)
7
8 print(f"Growth factor per day: {growth_factor:.3f}")
9 print(f"Percent growth per day: {percent_growth:.1f}%")
10 print(f"Estimated starting views: {starting_views:.1f} thousand")

```

This gives a growth factor of about 2.087 , a growth rate of 108.7 percent per day, and $4,027$ thousand views.

Exercise 1 Medication Concentration

Work

A patient takes a single dose of medication. A nurse measures the concentration remaining in the bloodstream every hour:

Hours (x)	0	1	2	3	4	5	6
Concentration, mg/L (y)	80.9	65.1	57.1	43.8	36.3	30.8	23.8

1. Find $\log_{10}(y)$ for each hour.
2. Regressing $\log_{10}(y)$ on x gives $\log_{10}(\hat{y}) = -0.0872x + 1.9108$, with $r = -0.9979$. Use this to write the exponential model $\hat{y} = a \cdot b^x$.
3. Predict the concentration at $x = 9$ hours. Is this interpolation or extrapolation?

Answer on Page 9

1.2 Power Models

Unlike exponential growth, certain curved relationships do not grow by a constant factor at every step. A marine biologist measures the length and weight of several fish from the same species:

Length, cm (x)	10	15	20	25	30	35	40
Weight, g (y)	15.7	53.9	135.5	244.0	461.4	668.0	1068.6

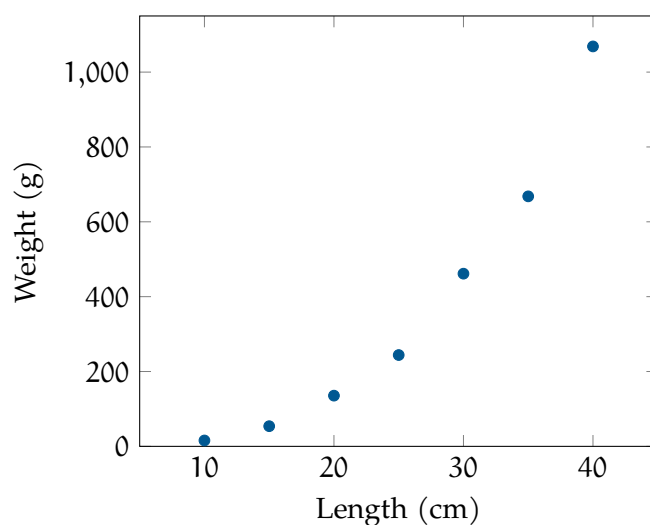


Figure 1.3: Fish weight against length. The curve bends upward, following a growth pattern distinct from exponential growth.

A straight line fit to this data yields $r^2 \approx 0.900$.

This kind of relationship follows a power model, $y = a \cdot x^b$. Take the logarithm of both sides:

$$\log(y) = \log(a \cdot x^b) = \log(a) + b \log(x)$$

Apply that to the fish data:

Length, cm (x)	10	15	20	25	30	35	40
Weight, g (y)	15.7	53.9	135.5	244.0	461.4	668.0	1068.6
$\log_{10}(x)$	1.000	1.176	1.301	1.398	1.477	1.544	1.602
$\log_{10}(y)$	1.196	1.732	2.132	2.387	2.664	2.825	3.029

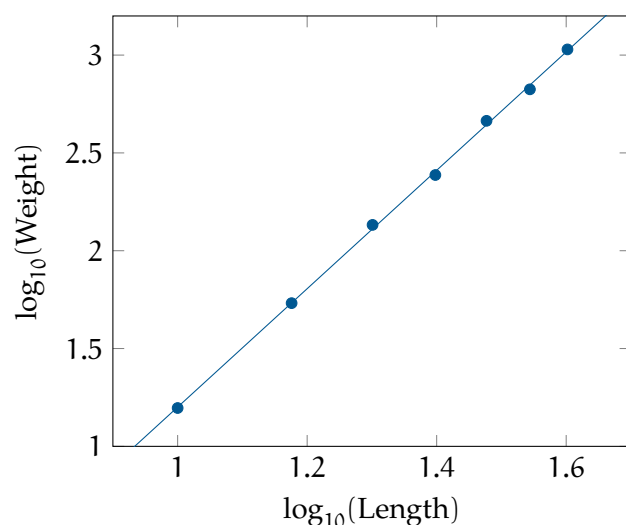


Figure 1.4: The fish data after a log-log transformation. Both axes are logarithms, and the points fall along a line.

Exercise 2 Fall Time

A student drops a ball from different heights and times how long it takes to hit the ground:

Height, m (x)	5	10	15	20	25	30	35
Fall time, s (y)	1.01	1.42	1.71	2.02	2.24	2.45	2.59

1. Explain why the shape of this scatterplot points to a power model rather than an exponential one.
2. Regressing $\log_{10}(y)$ on $\log_{10}(x)$ gives $\log_{10}(\hat{y}) = 0.4908 \log_{10}(x) - 0.3388$, with $r = 0.9996$. Use this to write the power model $\hat{y} = a \cdot x^b$.
3. Predict the fall time from a height of 50 m.

1.3 Choosing the Right Transformation

Both sections in this chapter start from the same move: something is curved, so take a logarithm and see if it straightens out. The two models call for a different logarithm, and mixing them up wastes time on a transformation that will not work.

Model	Equation	Transform	Linear in
Exponential	$y = a \cdot b^x$	$\log(y)$ vs. x	x
Power	$y = a \cdot x^b$	$\log(y)$ vs. $\log(x)$	$\log(x)$

This is a draft chapter from the Kontinua Project. Please see our website (<https://kontinua.org/>) for more details.

Answers to Exercises

Answer to Exercise 1 (on page 4)

Hours (x)	0	1	2	3	4	5	6
Concentration, mg/L (y)	80.9	65.1	57.1	43.8	36.3	30.8	23.8
$\log_{10}(y)$	1.908	1.814	1.757	1.642	1.560	1.489	1.377

Slope = $\log_{10}(b) = -0.0872$, so $b = 10^{-0.0872} \approx 0.818$. Intercept = $\log_{10}(a) = 1.9108$, so $a = 10^{1.9108} \approx 81.43$. The model is:

$$\hat{y} = 81.43 \cdot (0.818)^x$$

The concentration starts around 81.43 mg/L and multiplies by about 0.818 each hour, dropping roughly 18 percent per hour.

At $x = 9$: $\log_{10}(\hat{y}) = -0.0872(9) + 1.9108 = 1.126$, so $\hat{y} \approx 10^{1.126} \approx 13.37$ mg/L.

Answer to Exercise 2 (on page 6)

Slope = $b \approx 0.4908$. Intercept = $\log_{10}(a) = -0.3388$, so $a = 10^{-0.3388} \approx 0.4584$. The model is:

$$\hat{y} = 0.4584 \cdot x^{0.4908}$$

An exponent near 0.5 matches the physics: fall time under gravity is proportional to the square root of height, $t \propto \sqrt{h}$.

At $x = 50$: $\log_{10}(50) \approx 1.699$, so $\log_{10}(\hat{y}) = 0.4908(1.699) - 0.3388 = 0.495$, giving $\hat{y} \approx 10^{0.495} \approx 3.13$ s.



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